Crane Control Update and Sensor Calibration

Policy→PLC Interface, Tracking Accuracy, and Camera/Rack Calibration

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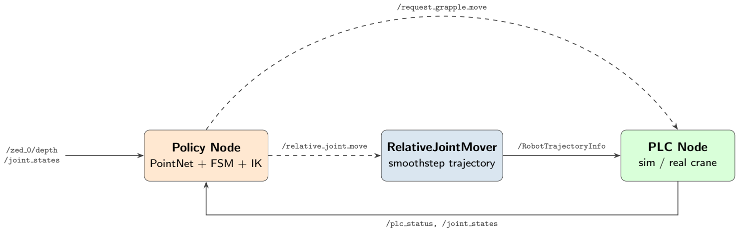
Outline

1. New control setup: policy $\rightarrow$ RelativeJointMover $\rightarrow$ PLC
2. Position-error characterization over the rack (5 targets)
3. Calibration: what we optimize (model-to-cloud ICP)
4. Adapting William's code for both cameras
5. Rack calibration: calibrated cameras + ArUco markers
6. Updating the sim assets / poses to match real
7. Next steps: train BC in sim, attempt zero-shot transfer

New Control Setup: Policy → RelativeJointMover → PLC



Policy Node (planning)

- Depth → 1024-pt base-frame cloud → policy → (x, y, z, ψ)
- FSM per phase: IK → joint *deltas* → `/relative_joint_move`; gripper → `/request_grapple_move`
- Advances on `/plc_status` (HOLD/EXECUTE/END)

RelativeJointMover (service)

- Relative deltas + velocity + min duration; target = current + delta
- **7th-order smoothstep** (zero end velocity & acceleration) at 50 Hz — hydraulic-friendly
- Publishes `RobotTrajectoryInfo`; same interface for sim & real

Position-Error Characterization (over the Rack)

Target (x, y, z)	ψ	**Task err (m)**
$(-4.0,\ 2.0,\ 1.5)$	0.00	0.014
$(-4.0,\ 2.0,\ 2.5)$	0.78	0.018
$(-3.5,\ 1.5,\ 2.0)$	-0.78	0.019
$(-3.5,\ 2.5,\ 2.0)$	1.57	0.012
$(-4.5,\ 2.0,\ 2.0)$	-1.57	0.010
	mean / max	**0.015 / 0.019**

Each target: IK $\rightarrow$ `/relative_joint_move`; error from FK of the achieved joints vs. commanded EE.

Result

Over the rack workspace the crane tracks to $\sim$**1.5 cm** (max 1.9 cm); per-joint error < 0.005 rad, yaw error $< 1.7°$.

Consistent Across the Rack

Accuracy holds across the swept height ($z = 1.5$–2.5 m) and the full yaw range ($\psi = \pm\pi/2$), well within the grasp tolerance.

Calibrating the Basemast Camera

The Mount Is Adjustable

The policy consumes a point cloud in the crane **base frame**, so the real cloud must land in the same frame as the sim cloud it trained on. The basemast camera sits on a **repositionable magnet bracket** (and can shift on it), so its pose on the mast is not fixed — it is **re-calibrated** directly to the mast whenever the mount moves.

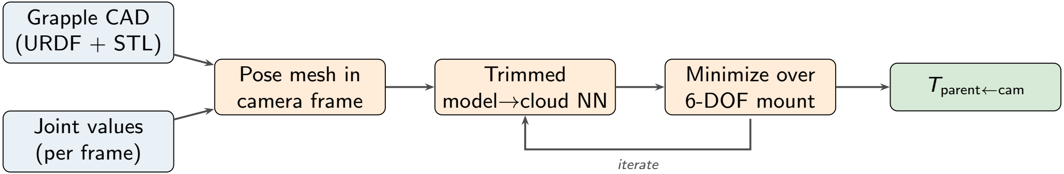
Sensors

- **zed_0** basemast (on the mast bracket, slews)
- **zed_1** stick (on the stick)
- **lidar_0** Ouster (same bracket; not used by the policy)

Use the Grapple

The **grapple** *does* move through the full workspace and is seen densely by both cameras — use it as the calibration object.

Grapple-Based Mount Calibration (Model-to-Cloud ICP)



The Cost

Pose the grapple mesh by FK, project into the camera, and minimize the **trimmed model-to-cloud** nearest-neighbour squared distance over the 6-DOF mount $(x, y, z, \text{roll}, \text{pitch}, \text{yaw})$. Model→cloud (not cloud→model): the cropped cloud has non-grapple structure that the reverse direction diverges on.

Accuracy Floor

RMS $\sim$47 mm (zed_1) / 64 mm (zed_0); higher for zed_0's longer chain (sag, encoder error).

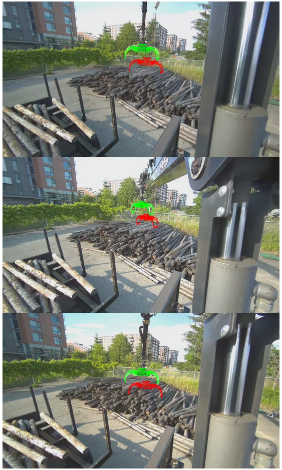
Calibrated Mounts (full 6-DOF)

$T_{\text{mast}\leftarrow\text{zed\_0}}$: xyz $(-0.168, -0.014, 0.897)$, rpy $(-106, -0.2, -1.9)^\circ$

$T_{\text{stick}\leftarrow\text{zed\_1}}$: xyz $(-0.133, 0.318, -0.009)$, rpy $(-90, 0.6, -2.3)^\circ$

large roll = camera optical-frame convention ($+Z$ forward)

Adapting William's Code for Both Cameras

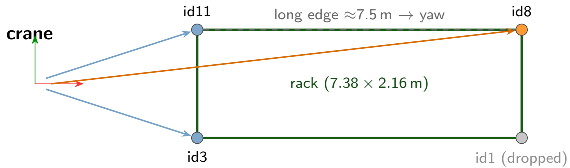


zed_0: posed grapple CAD (green) reprojected onto the observed grapple.

From William's render-and-compare

- Rebuilt against *our* URDF + STL meshes; FK validated to **0.0 mm / 0.00°** vs. bag `/tf`
- **zed_1** calibrates in `stick`, **zed_0** in `mast` (slew cancels in the cam→grapple chain)
- Dense grapple clouds (∼2300–3500 pts) from the motion bag (∼1132 cropped clouds/cam available); the fit uses ∼24 evenly-sampled poses

Rack Calibration: Fusing Both Cameras



- zed_0 (basemast): near posts
- zed_1 (stick): far post

Coverage

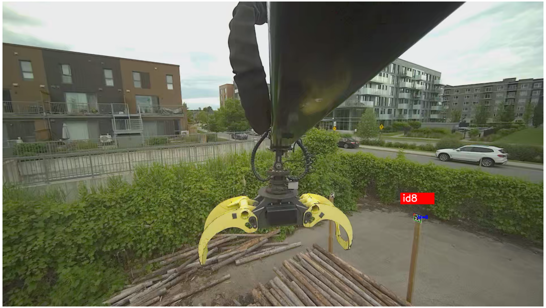
The rack is 7.38 m long; **no single camera** frames both ends with usable marker depth. Both cameras are calibrated into the **same base frame**, so markers seen by either one fuse into one rectangle.

Marker → Base

- Per marker: ArUco gives $T_{\text{cam}\leftarrow\text{marker}}$, lifted to base by FK · $T_{\text{parent}\leftarrow\text{cam}}$ (the **calibrated** mount)
- zed_0 → near posts `id3,id11`; zed_1 → far post `id8`. `id1` dropped (stick extrapolates poorly to that very-near 2 m marker, lands ∼1.3 m high)
- Cross-camera long edge `id11`→`id8` (≈7.5 m) → precise yaw 1.77° (vs. 2.34° from the short near edge)
- Post columns in the registered cloud fix corner XY (ArUco depth is the weak axis)

Rack Calibration: Marker Detections

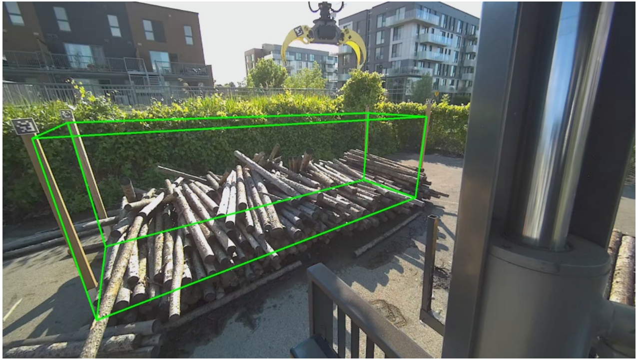
Basemast (zed_0): near posts **id3**, **id11**



Stick (zed_1): far post **id8** (grapple in foreground)

ArUco (DICT_6X6_50) detected in each camera's full-resolution frame, then lifted to the base frame through that camera's calibrated extrinsic. Together the views span the rack's long edge (`id11`→`id8`, ≈7.5 m), giving a precise yaw.

Rack Calibration: Result



Calibrated rack box (green) reprojected on the real basemast image; ArUco marker top-left.

Placed Rack

- True dims $2.16 \times 7.38 \times 2.4\,\mathrm{m}$
- Base pose: center $(-5.06, 2.53, -1.42)$, yaw 1.77°
- Bottom set **flat on the floor** (markers showed only a $\sim$1–2° tilt, flattened for sim)
- Action bounds set *inside* this box

Validation

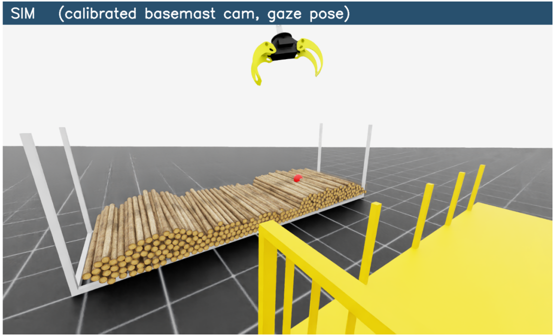
The marker-derived rack bottom matches the sim ground plane to **24 mm**, and the box edges follow the real posts in the image.

Updating the Sim to Match Real

Asset / Pose Updates

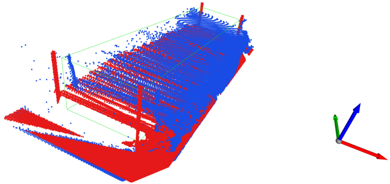
- Three sensor mounts baked into the **URDF** (full chain reproduces to machine precision)
- Basemast ZED + LiDAR frame added to the **sim USD**, pinned kinematically to the mast
- Rack placed at the calibrated base pose; logs and **action bounds** set in the base frame
- Gaze slew set so the sim basemast cam frames the rack

Sim vs. Real: Gaze View

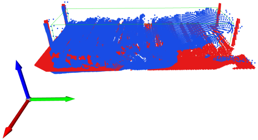


Basemast (zed_0) view at the gaze pose. The calibrated sim camera reproduces the real viewpoint: grapple top-center, log pile lower-left, rack frame, trailer right. This is the exact image the depth pipeline turns into the policy's point cloud.

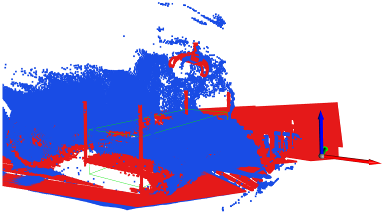
Sim vs. Real: Point Cloud (Base Frame)



Isometric



Front profile



Full scene (uncropped)

sim (red) vs. real (blue); action bounds (green). With the calibrated extrinsics the clouds land on the same rack in the base frame (shared extent $x \in [-6.9, -3.2]$, $y \in [-1.8, 6.8]$). The uncropped view shows the grapple and background that the action-bounds crop removes before inference. Remaining gap: real stereo has holes / noise and the floor sits $\sim$13 cm lower $\rightarrow$ depth-noise + point-dropout augmentation.

Next Steps

Plan

1. Rebuild the sim pile to the ordered layout; finalize reachable action bounds
2. **Train the BC policy in sim** on the calibrated gaze input
3. Attempt **zero-shot transfer** to the real crane (rosbag replay $\rightarrow$ dry run $\rightarrow$ live)
4. If transfer is poor: retrain with depth-noise / dropout / viewpoint augmentation

Questions?

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